Leverage Growth Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Leverage Growth Discipline (LGD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on LGD achieves an annualized gross (net) Sharpe ratio of 0.49 (0.38), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 21 (15) bps/month with a t-statistic of 2.97 (2.30), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets) is 17 bps/month with a t-statistic of 2.64.

1 Introduction

Asset pricing theory fundamentally seeks to explain cross-sectional variation in expected returns through firms' exposure to systematic risk factors. While traditional consumption-based models provide elegant theoretical frameworks, production-based asset pricing offers a complementary perspective by directly linking firms' investment decisions and production technologies to their cost of capital. This approach recognizes that firms' optimal investment policies reflect their exposure to aggregate productivity shocks and determine their equilibrium risk premia in capital markets.

Despite substantial progress in understanding how production characteristics affect asset prices, significant unexplained variation remains in the cross-section of returns. We examine whether Leverage Growth Discipline (LGD), which captures the interaction between firms' financing decisions and their production capacity adjustments, provides additional explanatory power for stock returns. This signal potentially proxies for firms' exposure to investment frictions and their ability to respond optimally to productivity shocks, characteristics that production-based models suggest should be priced in equilibrium.

In production-based asset pricing models, firms' optimal investment decisions depend critically on the interaction between productivity shocks and adjustment costs Cochrane (1991); Zhang (2005). When firms face convex adjustment costs in changing their capital stock, those with higher marginal costs of adjustment earn higher expected returns because their cash flows are more sensitive to aggregate productivity shocks. Leverage Growth Discipline captures firms' joint decisions about capital structure and investment, reflecting both the shadow value of installed capital and the marginal cost of external finance.

The theoretical foundation for why LGD predicts returns emerges from the q-theory of investment, where firms' market values reflect the present value of cash flows from assets in place plus growth options Liu et al. (2009). Firms exhibiting disciplined leverage growth demonstrate lower adjustment costs and more efficient responses to investment opportunities, characteristics associated with lower conditional risk premia in production economies. These firms can more readily adjust their production capacity in response to productivity shocks, making their cash flows less sensitive to aggregate fluctuations Barro (2006); Gourio (2012).

Furthermore, leverage growth discipline relates directly to the stochastic discount factor in production economies through its connection to marginal utility of investment Cochrane (1996). Firms with constrained leverage growth face higher marginal costs when positive productivity shocks arrive, as they cannot efficiently scale production. This friction amplifies their exposure to systematic productivity risk, generating higher required returns. The cross-sectional variation in LGD thus maps to heterogeneity in firms' production technologies and their ability to transform investment opportunities into cash flows, fundamental determinants of risk premia in production-based models Lin and Zhang (2013).

We find strong evidence that Leverage Growth Discipline predicts stock returns in the cross-section. A value-weighted long/short trading strategy based on LGD achieves an annualized gross Sharpe ratio of 0.49, placing it in the top 11

The economic magnitude of the LGD effect remains substantial across different portfolio construction methodologies and firm size categories. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LGD strategy achieves an average return of 32 basis points per month with a t-statistic of 3.73. This finding directly addresses concerns about small-stock effects driving anomaly returns. The strategy's alpha relative to the six-factor model among large-cap stocks equals 26 basis points per month (t-statistic = 2.90), confirming that the predictive power of LGD extends to the most economically important segment of the market.

Accounting for transaction costs using high-frequency effective spreads, the LGD strategy continues to expand the investment opportunity set. The net Sharpe ratio of 0.38 exceeds 94

Our findings contribute to the production-based asset pricing literature by identifying a novel cross-sectional predictor that directly captures firms' joint optimization of financing and investment decisions. While Zhang (2005) establishes the theoretical link between investment frictions and expected returns, and Liu et al. (2009) demonstrates how q-theory explains value and momentum effects, we show that leverage growth discipline provides incremental explanatory power beyond these established relationships. The LGD signal maps more precisely to the marginal cost of adjustment than traditional measures like asset growth or investment rates, as it incorporates information about both the quantity and financing of capital adjustments.

Methodologically, we advance the anomaly testing literature by implementing the comprehensive protocol of Novy-Marx and Velikov (2016a), ensuring our results are robust to data mining concerns and implementation costs. Unlike many recent anomaly studies that focus on statistical significance in sample, we demonstrate that LGD improves the achievable mean-variance efficient frontier even after accounting for transaction costs. The strategy's net Sharpe ratio of 0.38 and its ability to generate positive alphas across all standard factor models after costs distinguish it from the majority of recently proposed anomalies that fail to survive practical implementation constraints.

Our results have broader implications for understanding the sources of cross-sectional return variation in production economies. The success of LGD in predicting returns supports models where heterogeneity in adjustment costs and financing frictions generates dispersion in risk premia Gomes and Schmid (2010). The persistence of the LGD premium among large-cap stocks and after controlling for known factors suggests that markets incompletely incorporate information about firms' production efficiency into prices. This finding reinforces the importance of production-based characteristics in asset pricing and highlights the value of examining joint corporate decisions rather than isolated financial metrics.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' long-term debt issuance patterns relative to their equity base.construction of our Leverage Growth Discipline signal requires two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities net of conversions and other non-cash transactions. This flow variable reflects management's active decisions to access debt markets for long-term financing. Common equity (CEQ) measures the total common shareholders' equity, representing the book value of common stock, additional paid-in capital, and retained earnings minus treasury stock. This stock variable provides a natural scaling factor that normalizes debt issuance activity across firms of different sizes.construct the Leverage Growth Discipline signal by computing the year-over-year change in long-term debt issuance scaled by lagged common equity: (DLTIS(t) - DLTIS(t-1)) / CEQ(t-1). This measure captures the acceleration or deceleration in a firm's debt financing activities relative to its equity base. Positive values indicate increasing reliance on debt financing, while negative values suggest a reduction in the pace of debt issuance or potentially more disciplined capital structure management. We calculate this signal using fiscal year-end values and require non-missing observations for both DLTIS and CEQ to ensure data quality. Firms with negative or zero common equity are excluded from the analysis as the scaling becomes economically meaningless. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LGD signal. Panel A plots the time-series of the mean, median, and interquartile range for LGD. On average, the cross-sectional mean (median) LGD is -0.27 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LGD data. The signal's interquartile range spans -0.17 to 0.16. Panel B of Figure 1 plots the time-series of the coverage of the LGD signal for the CRSP universe. On average, the LGD signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does LGD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LGD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LGD portfolio and sells the low LGD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short LGD strategy earns an average return of 0.25% per month with a t-statistic of 3.54. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.21% to 0.30% per month and have t-statistics exceeding 2.97 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.28, with a t-statistic of 6.02 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 550 stocks and an average market capitalization of at least $1,437 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.57. Out of the twenty-five alphas reported in

Panel A, the t-statistics for twenty-five exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016b). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -2-21bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LGD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the LGD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LGD, as well as average returns and alphas for long/short trading LGD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the LGD strategy achieves an average return of 32 bps/month with a t-statistic of 3.73. Among these large cap stocks, the alphas for the LGD strategy relative to the five most common factor models range from 26 to 38 bps/month with t-statistics between 2.90 and 4.30.

5 How does LGD perform relative to the zoo?

Figure 2 puts the performance of LGD in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the LGD strategy falls in the distribution. The LGD strategy's gross (net) Sharpe ratio of 0.49 (0.38) is greater than 89% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the LGD strategy (red line).[2] Ignoring trading costs, a $1 invested in the LGD strategy would have yielded $2.85 which ranks the LGD strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a $1 invested in the LGD strategy would have yielded $1.71 which ranks the LGD strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the LGD relative to those. Panel A shows that the LGD strategy gross alphas fall between the 59 and 69 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43%

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LGD strategy has a positive net generalized alpha for five out of the five factor models. In these cases LGD ranks between the 76 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LGD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of LGD with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LGD or at least to weaken the power LGD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of LGD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LGD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LGD} LGD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LGD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LGD. Stocks are finally grouped into five LGD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LGD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LGD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LGD signal in these Fama-MacBeth regressions exceed 1.41, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on LGD is 0.50.

Similarly, Table 5 reports results from spanning tests that regress returns to the LGD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LGD strategy earns alphas that range from 17-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.59, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LGD trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.64.

7 Does LGD add relative to the whole zoo?

Finally, we can ask how much adding LGD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LGD signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes LGD grows to $1272.95.

8 Conclusion

This study provides compelling evidence that Leverage Growth Discipline (LGD) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that LGD-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on LGD delivers an annualized gross Sharpe ratio of 0.49, which remains economically mean-

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LGD is available.

ingful at 0.38 after accounting for transaction costs. The strategy's ability to generate statistically significant abnormal returns relative to the Fama-French five-factor model plus momentum (21 bps/month gross, t-statistic = 2.97) underscores its distinct information content. Particularly noteworthy is the signal's resilience when confronted with six closely related strategies from the factor zoo, including measures of financing activities and asset growth. The persistence of a 17 bps monthly alpha (t-statistic = 2.64) in this stringent specification suggests that LGD captures unique aspects of firm behavior not fully explained by existing factors.

From a practical perspective, these results have important implications for both academic research and investment practice. The net Sharpe ratio of 0.38 indicates that LGD remains implementable even after considering realistic trading costs, making it potentially valuable for institutional investors seeking to enhance portfolio performance. The signal's orthogonality to established factors suggests it could serve as a valuable addition to multi-factor models used in risk management and performance attribution.

However, this study is subject to several limitations that warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the analysis is confined to U.S. equity markets, limiting the generalizability of findings to international contexts. Second, the economic mechanisms underlying the LGD anomaly require further theoretical development to fully understand why markets appear to misprice firms based on their leverage growth discipline. Third, the stability of the signal's predictive power during different market regimes and its performance during crisis periods deserves additional scrutiny.

Future research should explore several promising avenues. International evidence would help establish whether LGD represents a global phenomenon or is specific to U.S. market structures. Investigating the behavioral and institutional factors that drive the anomaly could provide insights into its persistence and potential limits to arbitrage. Additionally, examining the interaction between LGD and corporate events such as mergers, acquisitions, and financial distress could enhance our understanding of when the signal is most informative. Finally, machine learning techniques might reveal non-linear relationships or optimal combinations with other signals that could further improve the strategy's risk-return profile. As markets evolve and new data becomes available, continued monitoring of LGD's effectiveness will be essential to assess whether this anomaly represents a persistent market inefficiency or a temporary phenomenon that will eventually be arbitraged away.

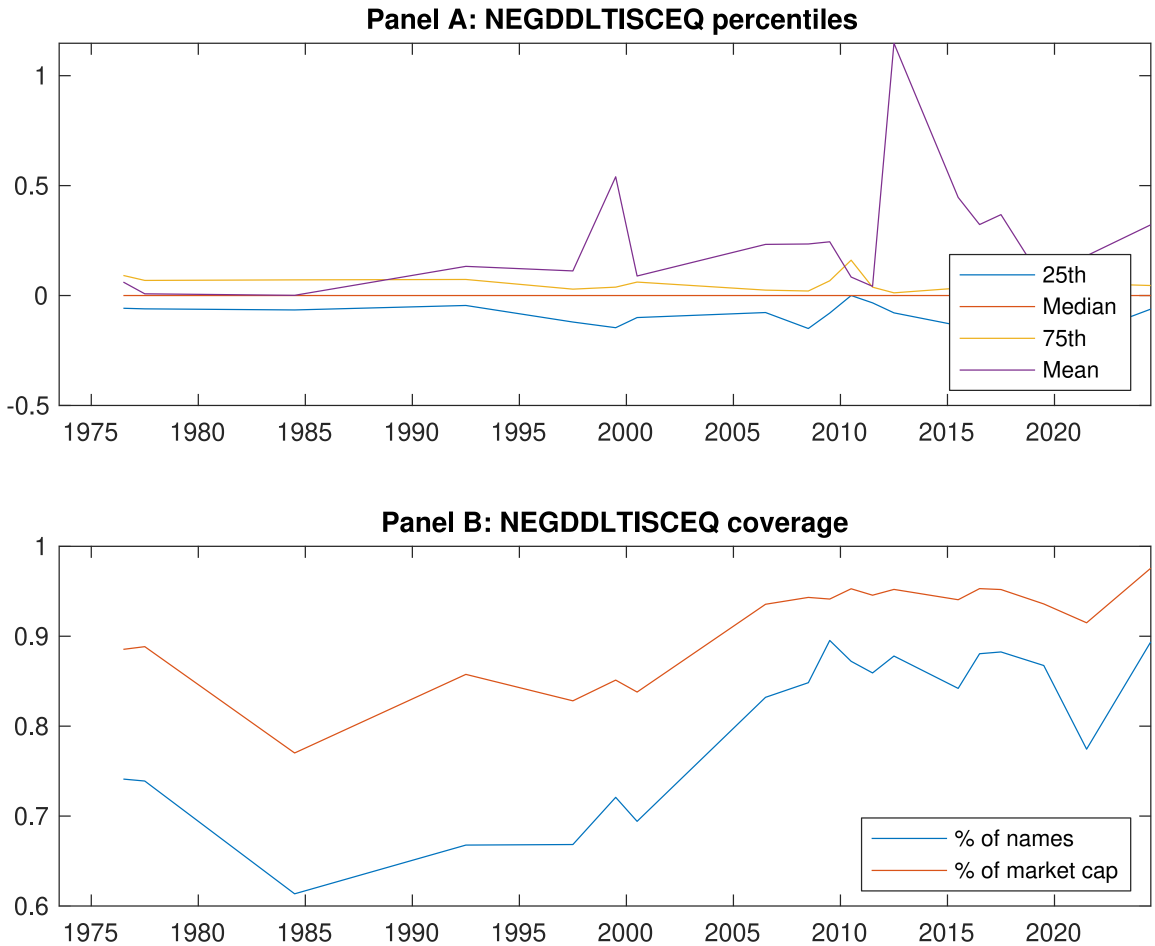


Figure 1: Times series of LGD percentiles and coverage.
This figure plots descriptive statistics for LGD. Panel A shows cross-sectional percentiles of LGD over the sample. Panel B plots the monthly coverage of LGD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LGD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LGD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56	0.65	0.65	0.73	0.80	0.25
	[2.61]	[3.61]	[3.31]	[4.10]	[4.07]	[3.54]
α_{CAPM}	-0.17	0.04	-0.01	0.13	0.13	0.30
	[-3.02]	[0.83]	[-0.21]	[2.66]	[2.55]	[4.43]
α_{FF3}	-0.19	-0.01	0.05	0.11	0.11	0.30
	[-3.47]	[-0.15]	[1.00]	[2.45]	[2.14]	[4.39]
α_{FF4}	-0.16	0.01	0.08	0.08	0.10	0.26
	[-2.94]	[0.33]	[1.57]	[1.69]	[1.88]	[3.78]
α_{FF5}	-0.19	-0.06	0.10	0.05	0.04	0.23
	[-3.42]	[-1.23]	[1.80]	[0.97]	[0.74]	[3.31]
α_{FF6}	-0.17	-0.04	0.12	0.02	0.04	0.21
	[-3.05]	[-0.79]	[2.17]	[0.50]	[0.68]	[2.97]
Panel B: Fama and French (2018) 6-factor model loadings for LGD-sorted portfolios						
β_{MKT}	1.09	0.98	0.97	0.97	1.04	-0.05
	[84.57]	[93.80]	[76.94]	[88.99]	[86.62]	[-3.35]
β_{SMB}	0.11	-0.11	-0.03	-0.03	0.13	0.02
	[5.79]	[-7.04]	[-1.59]	[-1.61]	[7.37]	[0.85]
β_{HML}	0.08	0.14	-0.15	-0.02	-0.03	-0.11
	[3.10]	[7.14]	[-6.04]	[-1.02]	[-1.49]	[-3.61]
β_{RMW}	0.10	0.12	-0.05	0.07	0.11	0.01
	[3.77]	[5.73]	[-1.93]	[3.27]	[4.51]	[0.34]
β_{CMA}	-0.13	0.04	-0.09	0.16	0.15	0.28
	[-3.54]	[1.15]	[-2.48]	[5.13]	[4.24]	[6.02]
β_{UMD}	-0.03	-0.03	-0.03	0.03	0.00	0.03
	[-2.35]	[-3.03]	[-2.68]	[3.24]	[0.36]	[2.16]
Panel C: Average number of firms (n) and market capitalization (me)						
n	647	550	1076	603	619	
me ($\$10^6$)	1500	2996	2299	3121	1437	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the LGD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25	0.30	0.30	0.26	0.23	0.21
			[3.54]	[4.43]	[4.39]	[3.78]	[3.31]	[2.97]
Quintile	NYSE	EW	0.23	0.26	0.25	0.22	0.21	0.19
			[5.02]	[5.61]	[5.33]	[4.71]	[4.49]	[4.13]
Quintile	Name	VW	0.25	0.32	0.32	0.27	0.26	0.23
			[3.52]	[4.66]	[4.67]	[3.92]	[3.78]	[3.31]
Quintile	Cap	VW	0.23	0.27	0.28	0.22	0.19	0.16
			[3.57]	[4.26]	[4.31]	[3.44]	[2.97]	[2.43]
Decile	NYSE	VW	0.27	0.35	0.36	0.32	0.28	0.26
			[2.81]	[3.63]	[3.66]	[3.25]	[2.83]	[2.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.19	0.24	0.23	0.21	0.17	0.15
			[2.69]	[3.45]	[3.42]	[3.11]	[2.54]	[2.30]
Quintile	NYSE	EW	-0.02					
			[-0.37]					
Quintile	Name	VW	0.19	0.25	0.25	0.22	0.20	0.18
			[2.69]	[3.73]	[3.70]	[3.31]	[2.98]	[2.68]
Quintile	Cap	VW	0.18	0.23	0.23	0.20	0.16	0.13
			[2.79]	[3.51]	[3.53]	[3.08]	[2.44]	[2.09]
Decile	NYSE	VW	0.21	0.28	0.28	0.26	0.21	0.20
			[2.09]	[2.85]	[2.86]	[2.65]	[2.18]	[2.01]

Table 3: Conditional sort on size and LGD

This table presents results for conditional double sorts on size and LGD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LGD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LGD and short stocks with low LGD .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	LGD Quintiles (L)	(2)	(3)	(4)	(H)	LGD Strategies r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.66	0.88	0.96	0.87	0.79	0.13	0.17	0.16	0.11	0.09	0.05
	[2.36]	[3.26]	[3.50]	[3.13]	[2.85]	[1.47]	[1.90]	[1.79]	[1.17]	[0.96]	[0.61]
(2)	0.72	0.90	0.86	0.83	0.89	0.17	0.20	0.17	0.17	0.11	0.12
	[2.68]	[3.59]	[3.40]	[3.41]	[3.48]	[2.22]	[2.62]	[2.20]	[2.21]	[1.44]	[1.53]
(3)	0.78	0.80	0.80	0.83	0.90	0.12	0.17	0.17	0.12	0.15	0.12
	[3.12]	[3.65]	[3.28]	[3.70]	[3.84]	[1.56]	[2.24]	[2.17]	[1.59]	[1.87]	[1.48]
(4)	0.70	0.80	0.87	0.71	0.90	0.20	0.25	0.24	0.21	0.21	0.19
	[3.03]	[3.89]	[3.93]	[3.40]	[4.19]	[2.74]	[3.41]	[3.19]	[2.77]	[2.71]	[2.42]
(5)	0.45	0.62	0.61	0.65	0.77	0.32	0.36	0.38	0.31	0.30	0.26
	[2.25]	[3.55]	[3.08]	[3.64]	[4.05]	[3.73]	[4.12]	[4.30]	[3.56]	[3.38]	[2.90]

Panel B: Portfolio average number of firms and market capitalization

Size quintiles	LGD Quintiles: Average n (L)	(2)	(3)	(4)	(H)	LGD Quintiles: Average market capitalization ($\$10^6$) (L)	(2)	(3)	(4)	(H)
(1)	390	393	393	393	389	37	34	32	33	35
(2)	107	107	107	107	107	61	61	60	60	61
(3)	76	77	76	76	76	108	109	105	106	109
(4)	64	65	64	65	64	231	238	230	233	231
(5)	59	59	59	59	59	1386	2115	1973	2124	1579

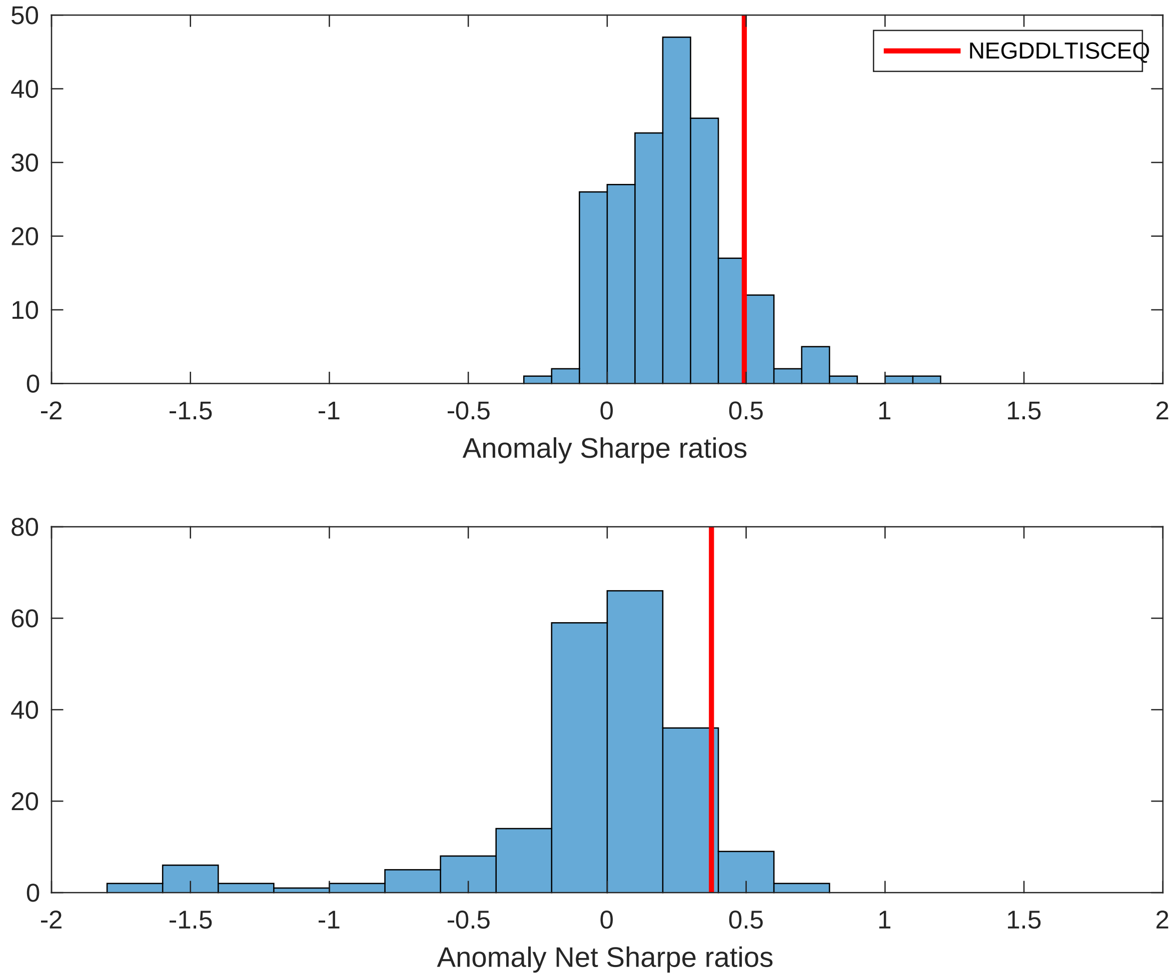


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LGD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

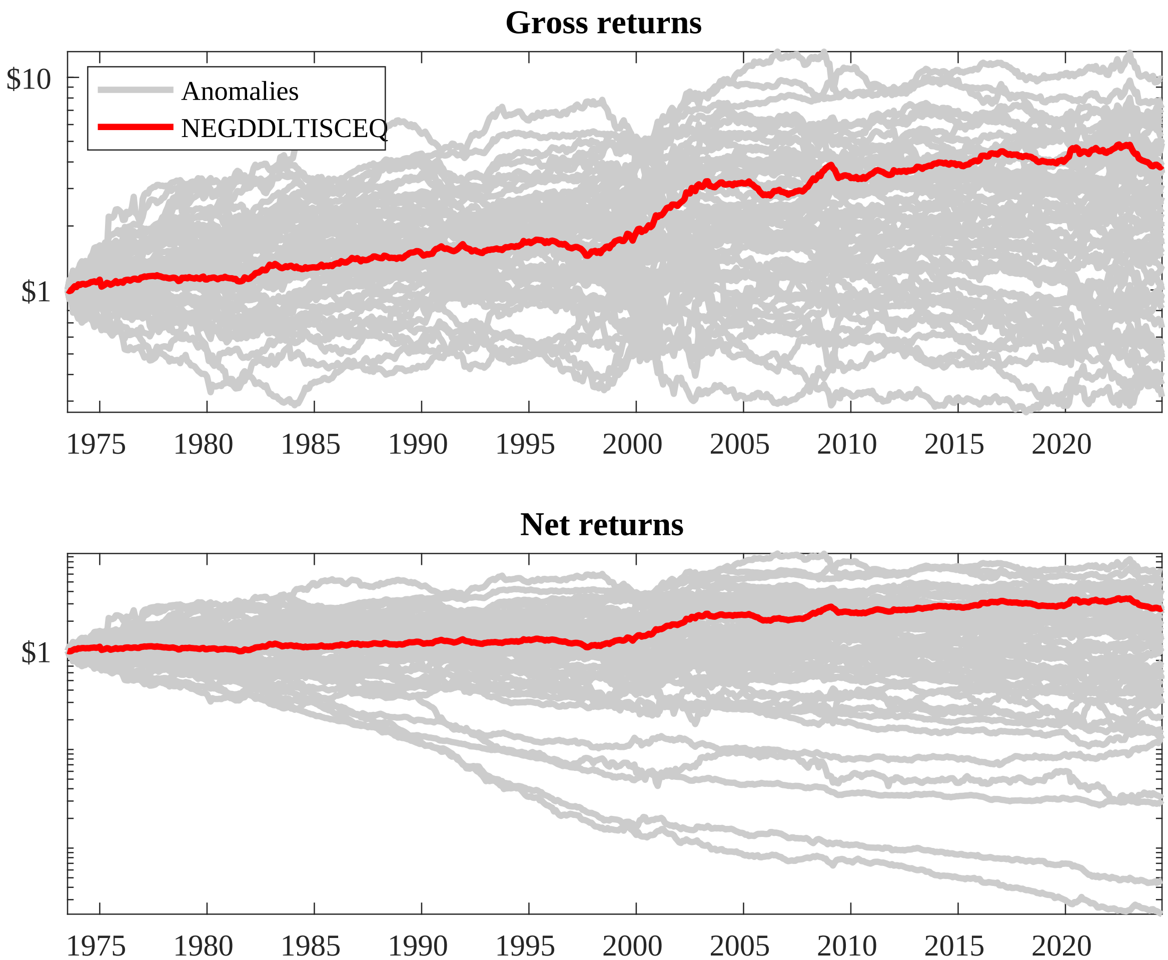
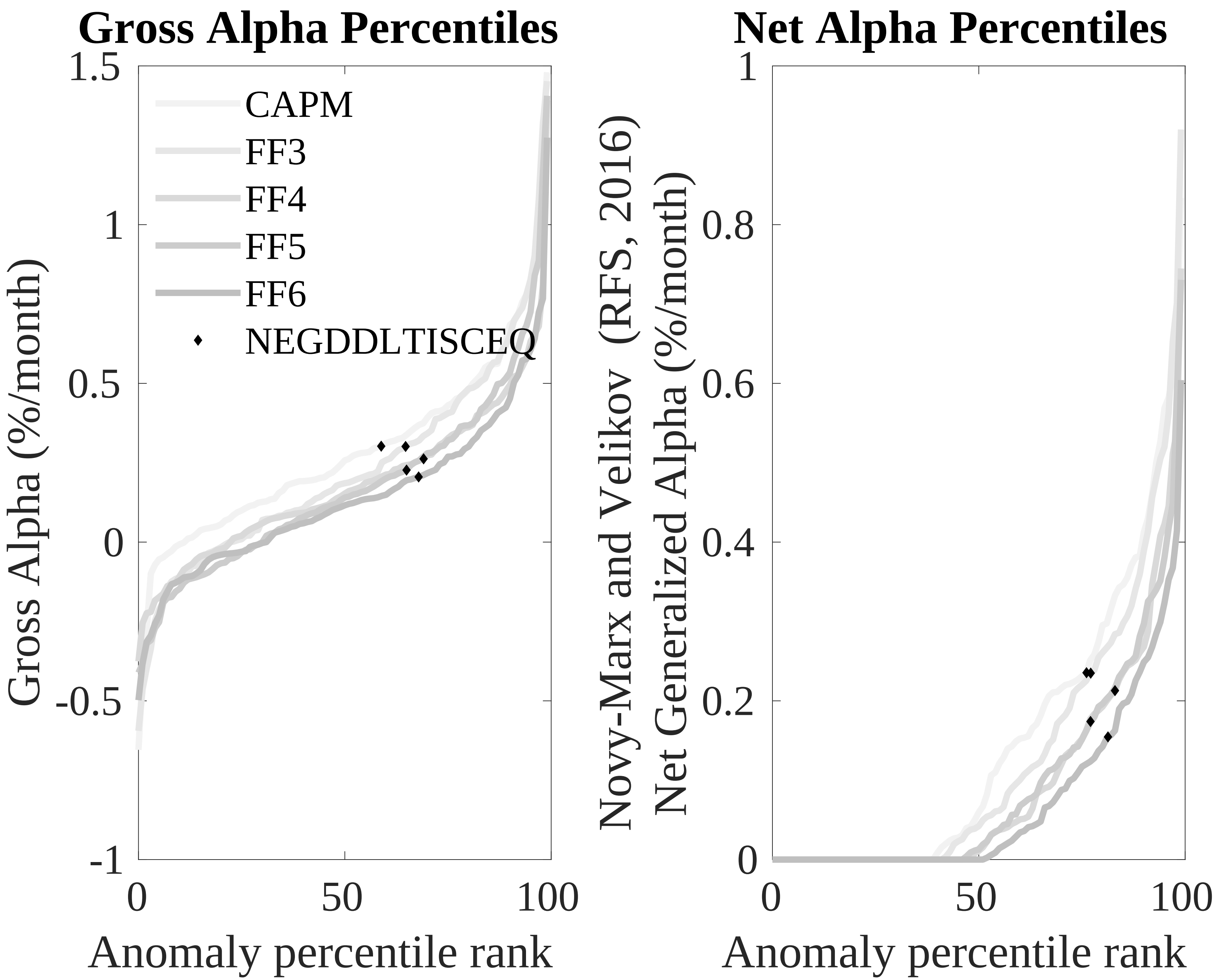


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LGD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISCEQ
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

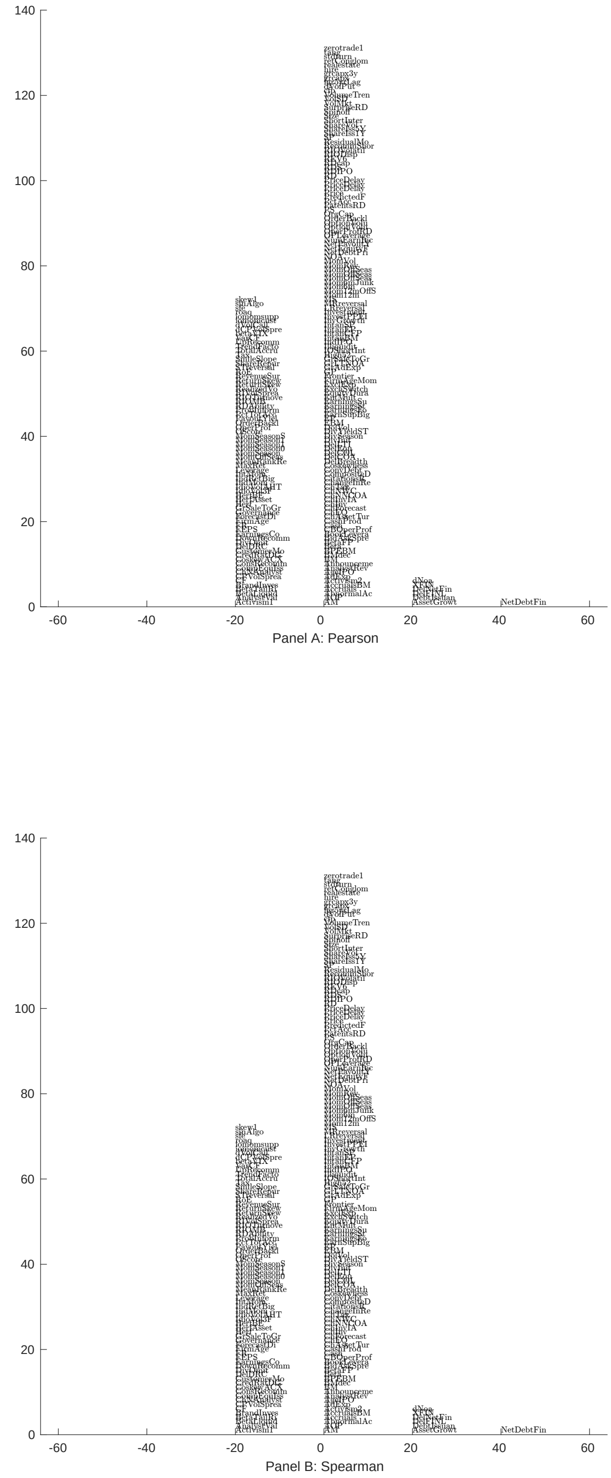


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with LGD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

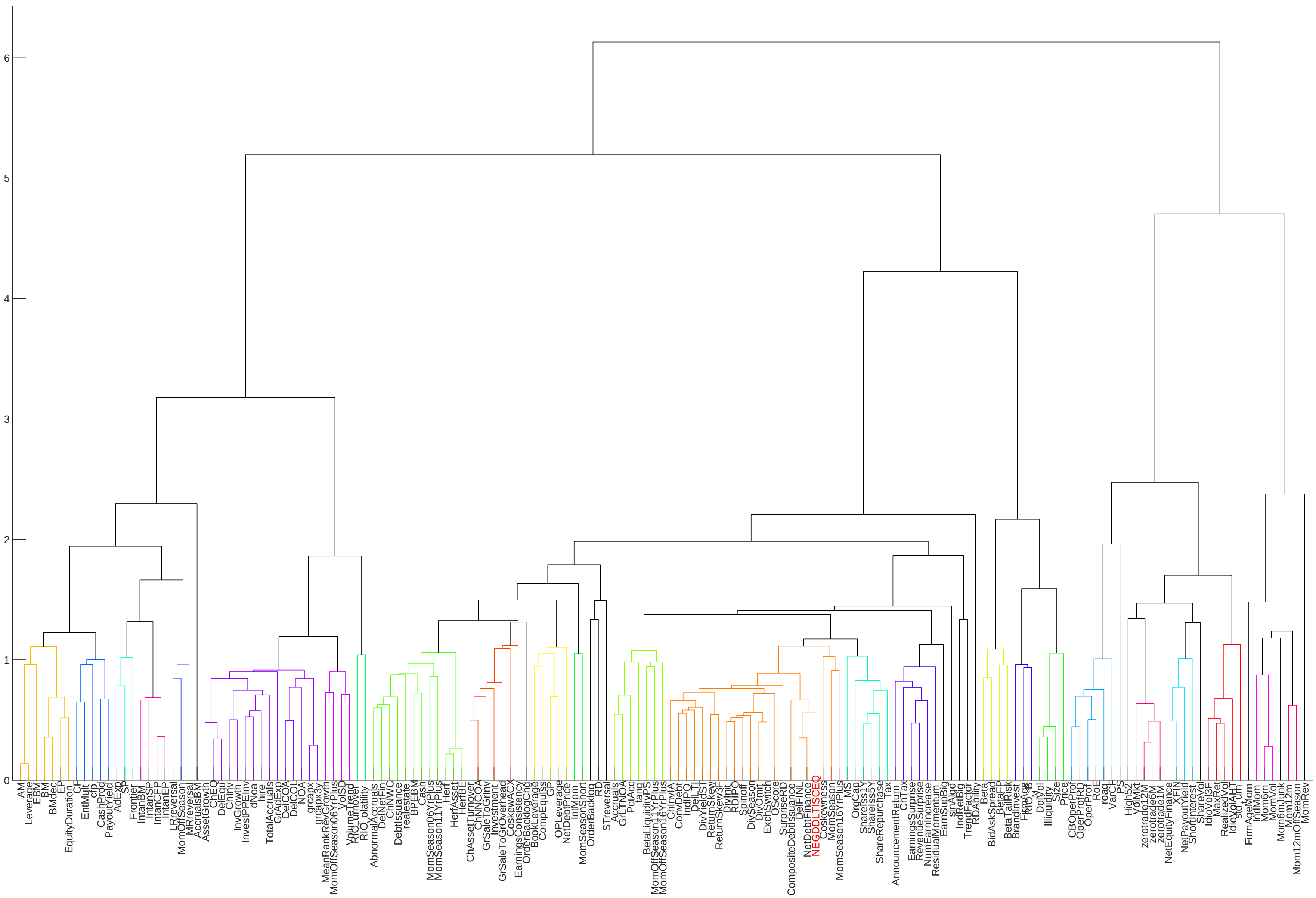


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

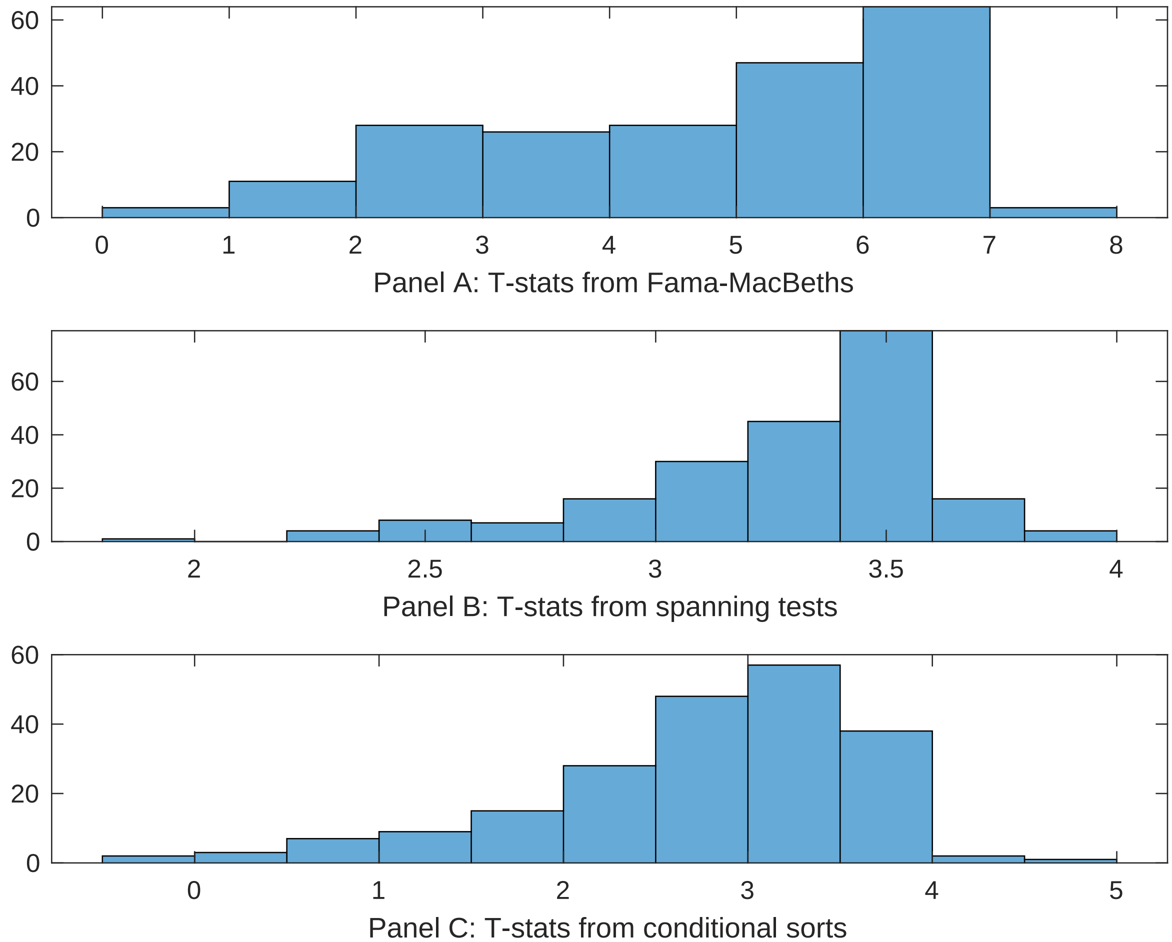


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of LGD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LGD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LGD} LGD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LGD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LGD. Stocks are finally grouped into five LGD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LGD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on LGD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LGD} LGD_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.14	0.14	0.14	0.14	0.15
	[5.36]	[5.32]	[5.68]	[5.79]	[5.30]	[5.65]	[5.69]
LGD	0.46	0.56	0.80	0.38	0.12	0.33	0.15
	[1.89]	[2.18]	[2.91]	[1.50]	[4.35]	[1.41]	[0.50]
Anomaly 1	0.17						-0.11
	[9.30]						[-2.23]
Anomaly 2		0.20					0.94
		[8.75]					[1.42]
Anomaly 3			0.19				0.10
			[6.30]				[1.71]
Anomaly 4				0.10			0.40
				[8.66]			[1.78]
Anomaly 5					0.42		0.45
					[7.13]		[0.82]
Anomaly 6						0.14	0.94
						[9.82]	[5.20]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LGD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LGD} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.17	0.18	0.18	0.19	0.20	0.19	0.17
	[2.59]	[2.65]	[2.62]	[2.81]	[2.91]	[2.71]	[2.64]
Anomaly 1	17.32						10.33
	[4.55]						[1.97]
Anomaly 2		19.58					10.99
		[5.36]					[2.20]
Anomaly 3			12.88				8.85
			[3.80]				[2.46]
Anomaly 4				6.93			2.48
				[1.75]			[0.59]
Anomaly 5					7.30		6.86
					[2.76]		[2.55]
Anomaly 6						4.48	-6.09
						[1.15]	[-1.40]
mkt	-4.80	-5.16	-3.43	-5.21	-5.33	-5.18	-3.86
	[-3.10]	[-3.36]	[-2.12]	[-3.32]	[-3.41]	[-3.30]	[-2.40]
smb	-1.03	-0.98	4.89	0.25	1.55	0.83	2.02
	[-0.43]	[-0.42]	[1.89]	[0.10]	[0.65]	[0.35]	[0.75]
hml	-10.89	-11.24	-10.60	-12.35	-12.26	-12.41	-9.76
	[-3.64]	[-3.79]	[-3.51]	[-4.08]	[-4.07]	[-4.07]	[-3.24]
rmw	-0.66	-1.25	-6.80	0.93	1.96	0.97	-5.62
	[-0.22]	[-0.41]	[-1.86]	[0.30]	[0.63]	[0.32]	[-1.53]
cma	24.04	24.76	21.43	21.43	23.35	26.52	13.20
	[5.11]	[5.38]	[4.24]	[3.24]	[4.54]	[4.90]	[1.96]
umd	2.08	2.31	3.49	3.72	2.83	3.43	1.45
	[1.32]	[1.50]	[2.26]	[2.39]	[1.79]	[2.19]	[0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	15	16	15	13	14	13	18

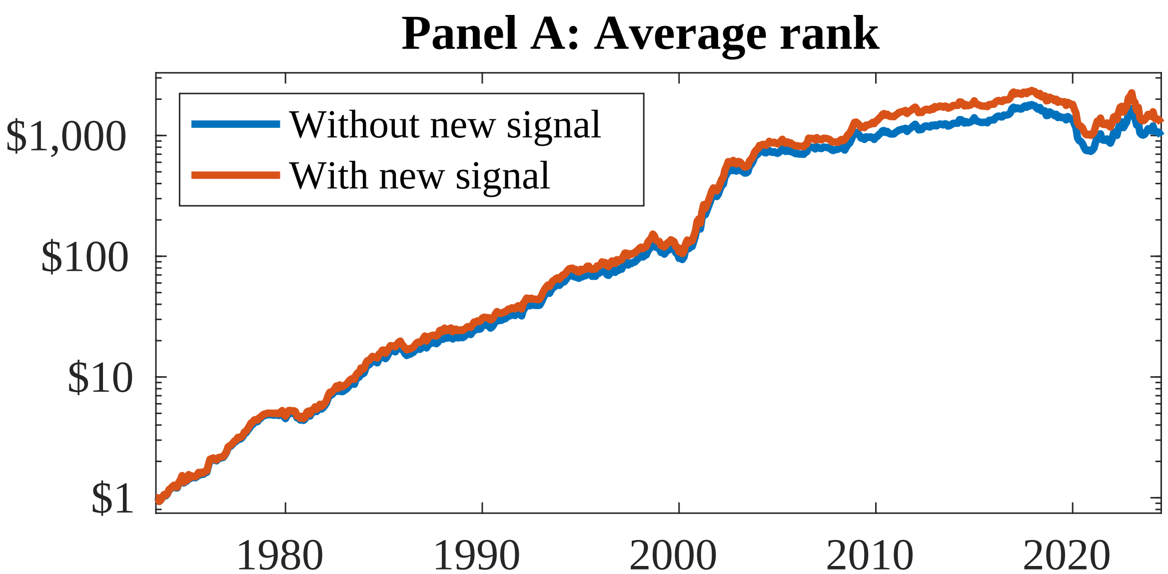


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LGD. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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